

Thoughts

A Quest for Perspective: Beyond the Arithmetic of Inequalities

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Daily Insight / Philosophy

L'inconnu qu'il s'agit de connaître m'apparaissait comme quelque étendue de terre ou de marne compacte, rétive à la pénétration... La mer s'avance insensiblement, sans bruit, rien ne semble se casser, rien ne bouge, l'eau est si loin qu'on l'entend à peine... pourtant elle finit par entourer la substance résistante.
— Récolte et Semailles

The unknown thing to be known appeared to me as some stretch of earth or hard marl, resisting penetration... the sea advances insensibly in silence, nothing seems to happen, nothing moves, the water is so far off you hardly hear it... yet it finally surrounds the resistant substance.

1. Some thoughts on equality and inequality: Hard and soft analysis

This log is a quest for vantage point when encountering inequalities. Rather than treating them as mere formulae and becoming mired into technical minutiae, the goal is to perceive the deeper structures that an inequality often partially conceals.

Hard analysts are accustomed to dealing with inequalities. They are not afraid of getting their hands dirty; their work involves performing estimations, tightening bounds and minimizing errors. To them, the game is in the precision of the estimate.

In contrast, soft analysts view inequalities as incomplete truths and feel compelled to uncover underlying equalities and isomorphisms. They operate on the belief that all inequalities stems from some incomplete observations of certain subtle structures. Mathematicians like Grothendieck, Atiyah and Hirzebruch etc. shared the conviction that inequalities are partial manifestations of an elegant, hidden equality. Ultimately, inequalities are able to be completed to an 'equality', perhaps by extending it into a long exact sequence, or considering discrepancies.

This ideology, which became deeply embedded in Algebraic Geometry and Topology from the 1950s to the 1970s, is illustrated by the following concrete examples:

Riemann-Roch

Tohoku Paper (1957)

People found that some functors are not exact for instance, the $\text{Hom}_R(-, N)$ functor, the tensor functor $(-) \otimes_R N$ and the section functor Γ . More precisely, given a short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, the sequence applied by Γ ,

$$0 \longrightarrow \Gamma(A, \mathcal{O}) \longrightarrow \Gamma(B, \mathcal{O}) \longrightarrow \Gamma(C, \mathcal{O}) \longrightarrow 0$$

is not exact at $\Gamma(C, \mathcal{O})$.

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2. Technical Lemma: Local Anatomy of Frey Curves

3. Computation Notes (Python/Mathematica)

```
1 # Finding torsion points on E
2 def check_torsion(a, b):
3     # Mathematica-style logic in Python
4     pass
```

To-Do List

- ☐ Finish the report on bilinear pairings (Tate/Weil).
- ☐ Update GitHub Pages with today's notes.